Vector Spaces and Subspaces

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Vector spaces generalize coordinate vectors. Subspaces are closed under linear combinations; column and null spaces record the action of a matrix, while bases and dimension measure independent directions.

Spaces and Subspaces

Subspace test:
$S \leq V$ iff $0 \in S$ and $u, v \in S$, $c, d \in \mathbb{R}$ imply $cu + dv \in S$.

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} = \text{span}\{a_1, \ldots, a_n\}, \quad \mathcal{N}(A) = \{x : Ax = 0\}.$$

$$\mathcal{R}(A) = \text{row}(A), \qquad \mathcal{N}(A^T) = \text{left nullspace}(A).$$

Basis theorem:
a basis is an independent spanning set. Every vector has a unique coordinate representation $v = \sum c_i q_i$.

$$\dim S = \text{number of vectors in any basis of } S.$$

Dimension theorem:
$\dim \mathcal{C}(A) = \text{rank}(A) = \dim \mathcal{R}(A)$.

Rank and Nullity

Rank-nullity theorem:
for $A : m \times n$, $\dim \mathcal{N}(A) + \text{rank}(A) = n$.

$$\dim \mathcal{C}(A) + \dim \mathcal{N}(A^T) = m.$$

Four-subspace dimension relations:
$\dim \mathcal{R}(A) = \dim \mathcal{C}(A) = r$, $\dim \mathcal{N}(A) = n - r$, $\dim \mathcal{N}(A^T) = m - r$.

Fundamental theorem:
$\mathcal{R}(A) \perp \mathcal{N}(A)$ and $\mathcal{C}(A) \perp \mathcal{N}(A^T)$.

$$\text{rank}(AB) \leq \min(\text{rank}\, A, \text{rank}\, B), \quad \text{rank}(A + B) \leq \text{rank}\, A + \text{rank}\, B.$$

PROBLEM SET

Q1.

(i) Determine which of the following subsets of $\mathbb{R}^3$ are subspaces. Justify each answer using the subspace requirements.

$$\begin{aligned} S_1 &= \{(x,y,z) : x+y+z=0\}, \\ S_2 &= \{(x,y,z) : x+y+z=1\}, \\ S_3 &= \{(x,y,z) : xyz=0\}, \\ S_4 &= \text{span}\{(1,4,0),(2,2,2)\}, \\ S_5 &= \{(x,y,z) : x \le y \le z\}. \end{aligned}$$

(ii) If S and T are subspaces of a vector space V, prove that $S \cap T$ is also a subspace of V.

Q2.

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, and let $[A\ b]$ denote the matrix obtained by appending b as a new column.

(i) Prove that

$$\mathcal{C}([A\ b]) = \mathcal{C}(A) \quad \Longleftrightarrow \quad b \in \mathcal{C}(A) \quad \Longleftrightarrow \quad Ax = b \text{ is solvable.}$$

(ii) Give one example in which appending b enlarges the column space and one example in which it does not.

Q3.

Consider the equation

$$x - 3y - z = 0.$$

(i) Write the equation as $Ax = 0$, identify the pivot and free variables, and find a basis for $\mathcal{N}(A)$.

(ii) Find the complete solution of $x - 3y - z = 12$ in the form $x_p + x_n$, where $x_n \in \mathcal{N}(A)$. Explain why this solution set is not a subspace.

Q4.

Consider the system

$$\begin{aligned} x + 2y - 2z &= b_1, \\ 2x + 5y - 4z &= b_2, \\ 4x + 9y - 8z &= b_3. \end{aligned}$$

(i) Use elimination on the augmented matrix to find the necessary and sufficient condition on b_1, b_2, b_3 for the system to be solvable.

(ii) When that condition holds, write the complete solution as one particular solution plus all solutions of the homogeneous system.

(iii) Identify the rank of the coefficient matrix and give a basis for its nullspace.

Q5.

Consider the two triples of vectors

$$\begin{aligned} u_1 &= (1,3,2), \quad & u_2 &= (2,1,3), \quad & u_3 &= (3,2,1), \\ v_1 &= (1,-3,2), \quad & v_2 &= (2,1,-3), \quad & v_3 &= (-3,2,1). \end{aligned}$$

Determine whether each triple is linearly independent. Hence determine which triples are bases for $\mathbb{R}^3$. Use elimination or a direct linear relation; do not use determinants.

Q6.

Let $\mathcal{P}_3$ be the vector space of real polynomials of degree at most 3.

(i) Give a basis for $\mathcal{P}_3$ and state its dimension.

(ii) Let $S = \{p \in \mathcal{P}_3 : p(1) = 0\}$. Prove that S is a subspace of $\mathcal{P}_3$, then find a basis and the dimension of S.

Q7.

For each matrix

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \end{bmatrix},$$

(i) Find a basis and the dimension of each of the four fundamental subspaces associated with A.
(ii) Find a basis and the dimension of each of the four fundamental subspaces associated with B.
(iii) For both matrices, verify the rank-nullity relations and the orthogonality of the corresponding row and null spaces.

Q8.

Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$.

(i) Prove that $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and hence that rank$(AB) \leq$ rank(A).
(ii) Prove that $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$. Use rank-nullity to deduce that rank$(AB) \leq$ rank(B).
(iii) For matrices $C, D \in \mathbb{R}^{m\times n}$, prove that

$$\text{rank}(C + D) \leq \text{rank}(C) + \text{rank}(D).$$

(iv) Give an example for which both bounds rank$(AB) \leq$ rank(A) and rank$(AB) \leq$ rank(B) are strict.